An inorganic compound (A) of S, Cl and oxygen has vapour density 67.5. It reacts with water to form two acids (B) and (C). (A) also reacts with KOH(aq) to forms two salts (D) and (E). (D) and (E) gives white precipitate with $AgNO_3$ and $BaCl_2$ solutions respectively. What are (A) to (E)?

$SOCl_2$, SO_2Cl_2

$32+16+71$ | $32+32+71 = 135$

$VD = 67.5$

$M = 2 \times 67.5 = 135$

$\underset{(A)}{SO_2Cl_2} + KOH(aq) \rightarrow \underset{(E)}{K_2SO_4} + \underset{(D)}{KCl} + H_2O$

$\underset{(A)}{SO_2Cl_2} + H_2O(l) \longrightarrow \underset{(B)}{H_2SO_4} + \underset{(C)}{HCl}$

$(E) \xrightarrow{BaCl_2} BaSO_4 \downarrow$

$(D) \xrightarrow{AgNO_3} AgCl \downarrow$ white

An acid (A) is pale-blue. The potassium salt of this acid does not give any reaction with $BaCl_2$ but gives white crystalline precipitate (B) with Ag+ ions. The acid (A) reacts with urea to liberate two gases (C) and (D). Gas (D) is used in synthesis of Urea also. On adding thiourea in acid (A) followed by addition of FeCl3/dilute HCl red coloured substance (E) is obtained. Identify substances (A) to (E).

$$\underset{(A)}{HNO_2} \xrightarrow{NH_2CONH_2} \underset{(C)}{N_2}\uparrow + \underset{(D)}{CO_2}\uparrow.$$

$$KNO_2 \xrightarrow{BaCl_2} \times$$

$$KNO_2 \xrightarrow{Ag^+} \underset{\text{white (B)}}{AgNO_2\downarrow}$$

$$HNO_2 \xrightarrow[\text{(Thiourea)}]{NH_2CSNH_2} SCN^- + N_2\uparrow$$

$$SCN^- \xrightarrow{FeCl_{3(aq)}} \underset{\text{Blood red (E)}}{[Fe(SCN)_3]_{(aq)}}$$

$A \longrightarrow HNO_2$

$B \longrightarrow AgNO_2$

$C \longrightarrow N_2$

$D \longrightarrow CO_2$

$E \longrightarrow [Fe(SCN)_3]$.

From the reactions given below, identify (A), (B), (C) and (D) and write their formulae.

$K_2Cr_2O_7$

(A)+dil. H_2SO_4 + (B) $\longrightarrow$ (B) Green Solution

(A) +dil. H_2SO_4 + (C) $\longrightarrow$ $\underline{MnSO_4}$

(A) $+O_2 \longrightarrow$ $\cancel{H_2O}$ (D)

$KMnO_4$ (O.A)

(D) $+BaCl_2 \longrightarrow$ White ppt.

A $\longrightarrow$ Should be a reducing agent.

$(\cancel{S^{2-}} / NO_2^- / \cancel{S^{2-}} / \underline{SO_3^{2-}}$ etc)

$\underset{(A)}{SO_3^{2-}} + H^+ + Cr_2O_7^{2-} \longrightarrow SO_4^{2-} + \underset{\text{green}}{Cr_2(SO_4)_{3(aq)}}$

$MnO_4^- + H^+ \longrightarrow Mn^{2+} + SO_4^{2-}$

$\underset{(s)}{Na_2SO_3} + \underset{(\text{atm. } O_2)}{O_2} \longrightarrow \underset{\text{or } Na_2SO_4}{SO_4^{2-}} \xrightarrow{BaCl_2} BaSO_4\downarrow$, White.

A waxy crystalline solid (A) with a garlic odour is obtained by burning white P in a stream of air and nitrogen. (A) reacts vigorously with hot water forming a gas (B) and an acid (C). Gas (B) has unpleasant odour of rotten fish and is neutral towards litmus. When passed through $AgNO_3$ solution, gas (B) produces a black precipitate (D). What are (A) to (D)? Give chemical equations of the reactions.

Sol^n:

$$P_4 + \underset{(\underset{21\%}{O_2} + \underset{78\%}{N_2})}{\text{air}} \longrightarrow \underset{(A)}{\overset{+3}{P_4O_6}} \xrightarrow[\text{(disproportionation)}]{+H_2O(l)} \underset{(B)}{PH_3}\uparrow + \underset{(C)}{H_3PO_4}$$

$$PH_3 \xrightarrow{AgNO_3\ (OA)} \underset{(D)}{Ag}\downarrow + H_3PO_3$$

An orange solid (A) on heating gives a green residue (B), a colourless gas (C) and water vapour. The dry gas (C) on passing over heated Mg gave a white solid (D). (D) on reaction with water gave a gas (E) which formed dense white fumes with HCl. Identify (A) to (E) giving reactions.

NH_3

Sol^n:

$$\underset{\substack{\text{orange solid}\\(A)}}{(NH_4)_2Cr_2O_7(s)} \xrightarrow{\Delta} \underset{(C)}{N_2\uparrow} + \underset{\substack{\text{green}\\(B)}}{Cr_2O_3(s)} + H_2O\uparrow$$

$$N_2 \xrightarrow[\Delta]{+Mg} \underset{(D)}{Mg_3N_2}$$

$$Mg_3N_2 \xrightarrow{+H_2O(l)} Mg(OH)_2\downarrow + \underset{(E)}{NH_3}\uparrow$$

$$NH_3 \xrightarrow{+HCl} \underset{\text{white fumes}}{NH_4Cl}\uparrow$$

Match the reaction products listed in column–I with the particulars listed in column-II

Column–I			Column–II
(A) $XeF_2 + H_2O \longrightarrow$	p, r, s, t	(p)	Redox reaction
(B) $XeF_4 + H_2O \longrightarrow$	p, q, r, s, t	(q)	Disproportionation
(C) $XeF_6 + H_2O \longrightarrow$	t	(r)	O_2 formation
(D) $XeO_3 + NaOH \longrightarrow$	p, q, r, s	(s)	Xe formation
		(t)	Etching glass

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$$XeF_2 + H_2O(l) \longrightarrow Xe\uparrow + O_2\uparrow + HF(aq) \xrightarrow{+SiO_2} SiF_4\uparrow + H_2O$$

$$SiF_4 \xrightarrow{+HF} H_2SiF_6$$

$$XeF_4 + H_2O(l) \longrightarrow \underset{0}{Xe}\uparrow + \underset{+6}{XeO_3} + O_2\uparrow + HF(aq)$$

$$XeF_6 + H_2O(l) \longrightarrow XeO_3 + HF$$

$$\underset{+6}{XeO_3} + NaOH \longrightarrow NaHXeO_4 \xrightarrow{OH^-} \overset{0}{Xe}\uparrow + O_2\uparrow + \overset{+8}{XeO_6^{4-}}(aq)$$

(JEE-Adv)

(perxenate)

$x - 12 = -4$

$x = +8$

Match the reactions listed in column-I with the product(s) listed in column-II.

Column - I		Column - II	
(A) $Cl_2O_6 + H_2O \longrightarrow$	q, s	(p)	ClO_2
(B) $NaClO4(s) + HCl(conc.) \longrightarrow$	s	(q)	$HClO_3$
(C) $KClO_3 + (COOH)_2 \longrightarrow$	p	(r)	Cl_2O
(D) $HgO + Cl_2 \xrightarrow{573K}$	r	(s)	$HClO_4$

$HClO_4$ (+7)

$HClO_3$ (+5)

$HClO_2$ (+3)

$HClO$ (+1)

Sol^n:

$\underset{(+6)}{Cl_2O_6} + H_2O(l) \longrightarrow HClO_3 + HClO_4$

(mixed anhydride)

$NaClO_4 + HCl(conc) \longrightarrow HClO_4 + NaCl$

$\underset{(OA)}{KClO_3} + \underset{(RA)}{H_2C_2O_4} \longrightarrow \underline{ClO_2} + CO_2\uparrow$

$HgO + Cl_2 \xrightarrow{\Delta} Cl_2O + HgCl_2$

PH$_3$ can be obtained by

(A) heating hypophosphorus acid

(B) heating orthophosphorus acid

(C) reacting white phosphorus with hot conc. NaOH

(D) Hydrolysis of calcium phosphide

Sol^n: (A) $H_3PO_2 \xrightarrow[\Delta]{t_1\,^\circ C} H_3PO_3 + PH_3\uparrow$

hypophosphorous acid

$H_3PO_3 \xrightarrow{t_2\,^\circ C} H_3PO_4 + PH_3\uparrow$

$(t_1 < t_2)$

Disproportionation Rex.

(B) $H_3PO_3 \xrightarrow{\Delta} H_3PO_4 + PH_3\uparrow$

(C) $P_4 + NaOH_{(aq)} \xrightarrow[\Delta]{+H_2O(l)} PH_3\uparrow + NaH_2PO_{2(aq)}$.

(D) $Ca_3P_2 + H_2O_{(l)} \rightarrow Ca(OH)_2 + PH_3\uparrow$

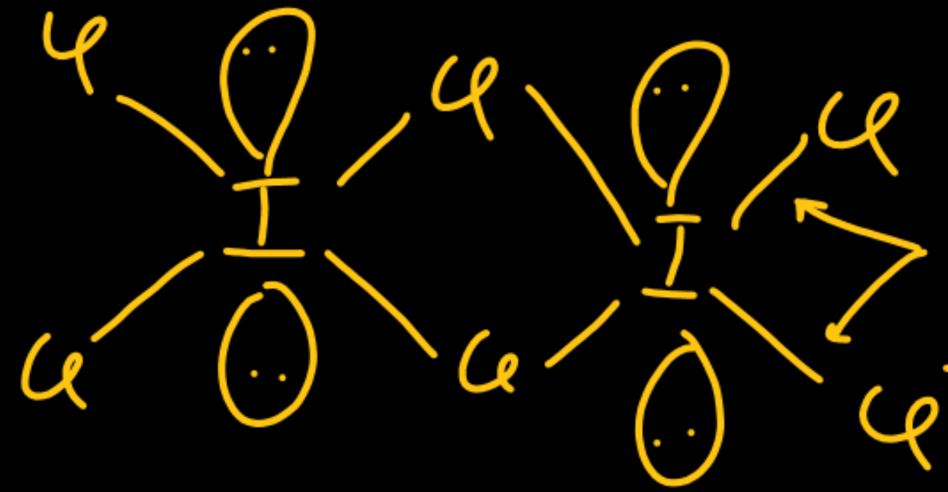
* Pure PH$_3$ is non-inflamable like NH$_3$.

Impure PH$_3$ has vapours of P_2H_4 and P_4 (inflamable)

In which of the following compound(s) terminal (2C – 2e⁻) bond and bridge bonds are lying in same plane:

(A) I_2Cl_6 (B) Fe_2Cl_4 (C) Solid $BeCl_2$ (D) Ga_2H_6

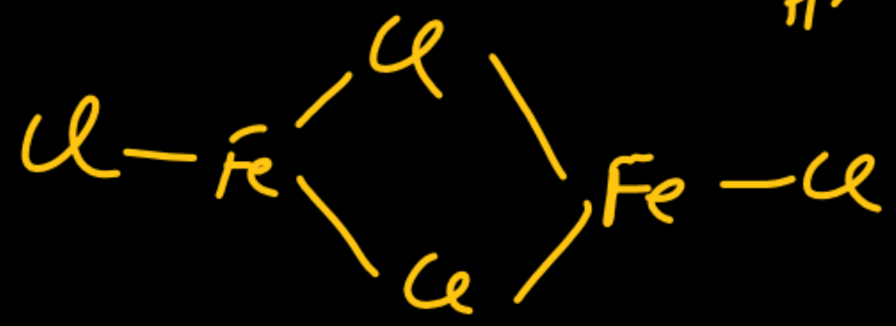
Sol^n:



terminal

Planar molecule

Fe_2Cl_4



Planar

The species which react with silica/glass in presence of moisture is.

(A) HF (B) XeF_2 (C) XeF_4 (D) XeF_6

$SiO_2 + HF \longrightarrow SiF_4 + H_2O$

$SiF_4 \xrightarrow{+HF} H_2SiF_6$.

HF: used to etch glass.

H_2SiF_6: fluorosilicic acid

$XeF_2 + H_2O(l) \longrightarrow Xe + O_2 + HF$

$Xe + O_2 + HF \xrightarrow{SiO_2} H_2SiF_6$

$XeF_4 + H_2O \longrightarrow XeO_3 + Xe + O_2 + HF$

$HF \xrightarrow{+SiO_2} H_2SiF_6$.

$XeF_6 + SiO_2 \longrightarrow XeOF_4 + SiF_4$

$XeOF_4 \xrightarrow{+SiO_2} XeO_2F_2 + SiF_4$

$XeO_2F_2 \xrightarrow{+SiO_2} XeO_3 + SiF_4$.

Oxygen is not evolved when:

✗ (A) potassium chlorate is heated with MnO_2 catalyst

✗ (B) sodium peroxide reacts with warm water

✓ (C) ammonium nitrate is heated

✓ (D) zinc oxide is treated with NaOH

(A) $KClO_{3(s)} \xrightarrow[\Delta]{MnO_2} KCl + O_2\uparrow$

(B) $Na_2O_2 + H_2O_{(l)} \longrightarrow NaOH + H_2O_2$

$H_2O_2 \xrightarrow{\Delta} H_2O + O_2\uparrow$

(C) $NH_4NO_3 \xrightarrow{\Delta} N_2O\uparrow + H_2O$

(D) $ZnO + NaOH_{(aq)} \longrightarrow Na_2ZnO_{2(aq)} + H_2O$

$O_2^{2-} + H_2O \longrightarrow H_2O_2$

$H_2O_2 \xrightarrow{\Delta} H_2O + O_2\uparrow$

$O_3^{-} + H_2O \longrightarrow H_2O_2 + O_2\uparrow$

Which statements are correct for ozone?

✓ (A) It is obtained by silent electric discharge on oxygen

✓ (B) It has both O–O bonds of equal bond length

✓ (C) It is regarded as an allotrope of oxygen

✗ (D) Ozone molecules is paramagnetic like oxygen molecule

A, B, C

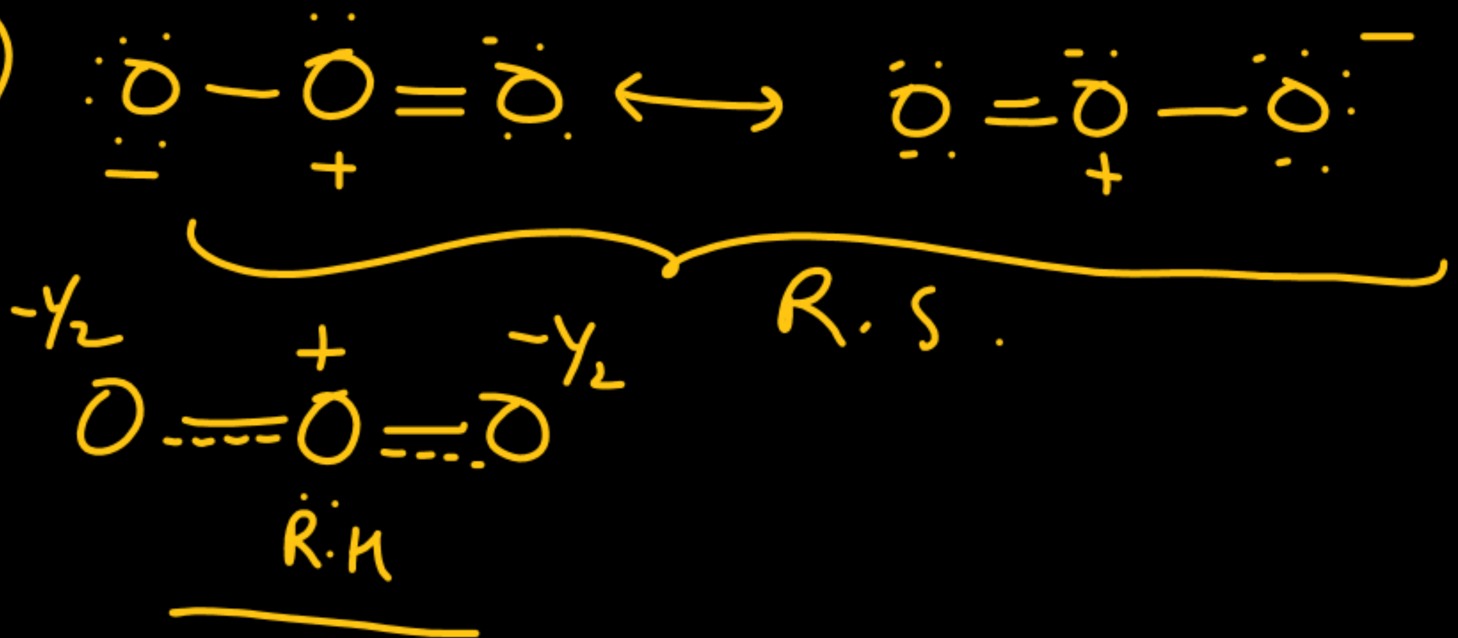
(A)

$$O_2 \xrightarrow[\text{discharge}]{\text{electrical}} O + O$$

$$O_2 + O \longrightarrow O_3$$

(we get ozonised oxygen)

(B)



White crystalline solid (A) reacts with H_2 to form a highly associated liquid (B) and a monoatomic, colorless gas (C). The liquid (B) is used for etching glass. Compound (A) undergoes hydrolysis slowly to form (C), (B) and a diatomic gas (D) whose IE is almost similar to that of (C). (B) forms an addition compound with KF to form (E) which is electrolysed in the molten state to form a most reactive gas (F) which combines with (C) in 2:1 ratio to produce (A).

$B \rightarrow HF$, $C \rightarrow Xe$

$IE_{Xe} \cong IE_{O_2}$

$A \rightarrow \underbrace{XeF_2, \cancel{XeF_4}}$

$\downarrow + H_2O(l)$

$Xe + HF + \underset{(D)}{O_2}$

$HF + KF \rightarrow \underset{(E)}{KHF_2}\ (l)$

electrolysis

$\underset{(F)}{F_2} + Xe \rightarrow \underset{(A)}{XeF_2}$

2 : 1

According to Molecular Orbital Theory, which of the following is correct about the molecule D?

(A) its bond order is 2.0

(B) it has two unpaired electrons in π−bonding M.O. (π^*)

(C) both the above are correct

(D) none of these is correct

Which of the following is correct for the white crystalline solid (A) ? → XeF_2

(A) It oxidises F^- to F_2

(B) It on hydrolysis with alkali undergoes disproportionation.

(C) It is obtained by the reaction of (C) with O_2F_2 at 118°C.

(D) None of these.

$Xe + O_2F_2 \rightarrow XeF_2 + O_2$

The compound 'A' reacts with sulphur to form a compound in which hybridisation state of sulphur atom is

$S + \underset{(OA)}{XeF_2} \rightarrow SF_6 + Xe$

(A) sp^3d (B) sp^3d^2 (C) sp^3 (D) sp^3d^3